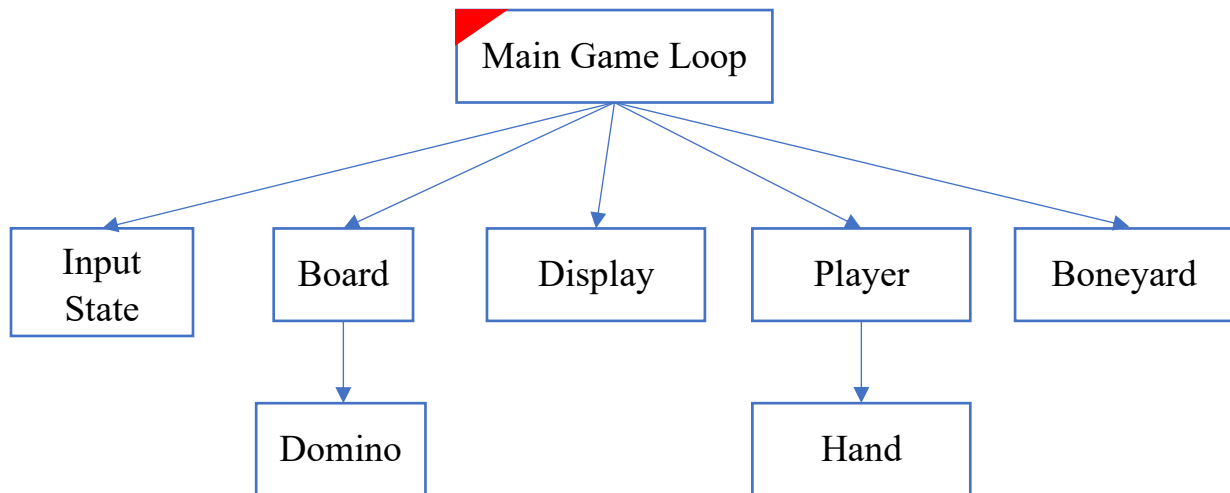


Proposed Design



Trigger: update from mouse

Description of the Design

Main Game Loop: Central manager of the game flow and state

- Data:
 - currentPlayer: Reference to the Player whose turn it is.
 - Board: Reference to the Board object.
 - Boneyard: Reference to the Boneyard object.
 - Display: Reference to the Display object.
- Behavior:
 - startGame(): Initializes the game, shuffles and distributes dominos.
 - playTurn(): Manages the sequence of actions in a player's turn.
 - Endgame(): Handles game end conditions and determines the winner.

Input State: Manages and validates user input.

- Behavior:
 - playerMovement(): Requests and receives the action chosen by the player.
 - validateMove(Domino, Position): Check if the move is valid.

Board: Contains the dominos, represents the playing area.

- Data:
 - dominos: List of Domino objects that show on the board.
- Behavior:
 - addDomino(Domino): Place a new domino on the board.
 - isValidMove(Domino, Position): Checks if a domino can be placed at a given position.

Player: Represents a player in the game.

- Data:
 - hand: Reference to the player's Hand object.
 - Score: Integer representing the player's current score.
- Behavior:
 - playDomino(Domino): Removes a domino from the hand and places it on the board.
 - drawBoneyard(): Takes a domino from the boneyard and adds it to the hand.
 - calculateScore(): Computes the player's score at the end of the game.

Boneyard: Represents the pool of undrawn dominos.

- Data:
 - dominos: List of Domino objects not yet in play.
- Behavior:
 - shuffle(): Randomizes the order of dominos in the boneyard.
 - drawDomino(): Removes and returns the top domino from the boneyard.
 - isEmpty(): Checks if there are any dominos left in the boneyard.

Display: Handles the visual representation of the game state.

- Behavior:
 - updateBoard(Board): Refreshes the display of dominos on the board.
 - updatePlayerHand(Hand): Shows the current player's available dominos.
 - showMessage(String): Displays game messages, prompts, or results to the player.

Domino: Represents a single domino piece.

- Data:
 - leftValue: Integer for the left side of the domino.
 - rightValue: Integer for the right side of the domino.
- Behavior:
 - rotate(): Swaps the left and right values.
 - matches(Domino): Checks if this domino can connect with another domino.

Hand: Manages the set of dominos held by a player.

- Data:
 - dominos: List of Domino objects in the player's possession.
- Behavior:

- addDomino(Domino): Adds a domino to the hand.
- removeDomino(Domino): Removes a specific domino from the hand.
- hasDomino(Domino): Checks if a specific domino is in the hand.